IDS Chaperone Drug Design

From Clinical Data to Thermodynamic Pharmacophore

Targeting the C422F Neo-Pocket

Slide 1: The Clinical Illusion vs Thermodynamic Reality

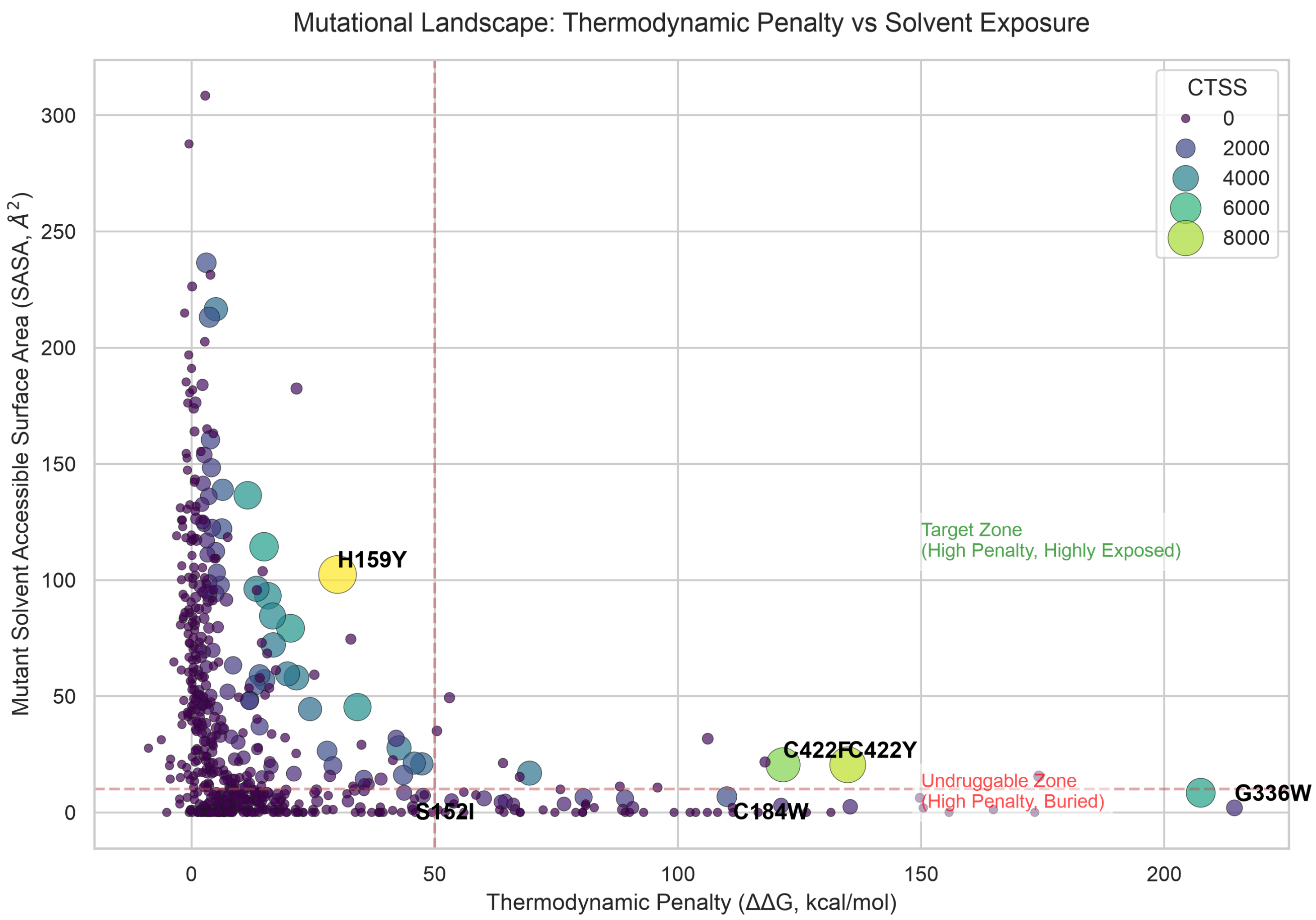
Problem Statement:

- Traditional target S152I is clinically frequent but lacks structural destruction.
- High frequency does not equal high druggability.
- AlphaFold predictions often show intact cores for catastrophic mutants.

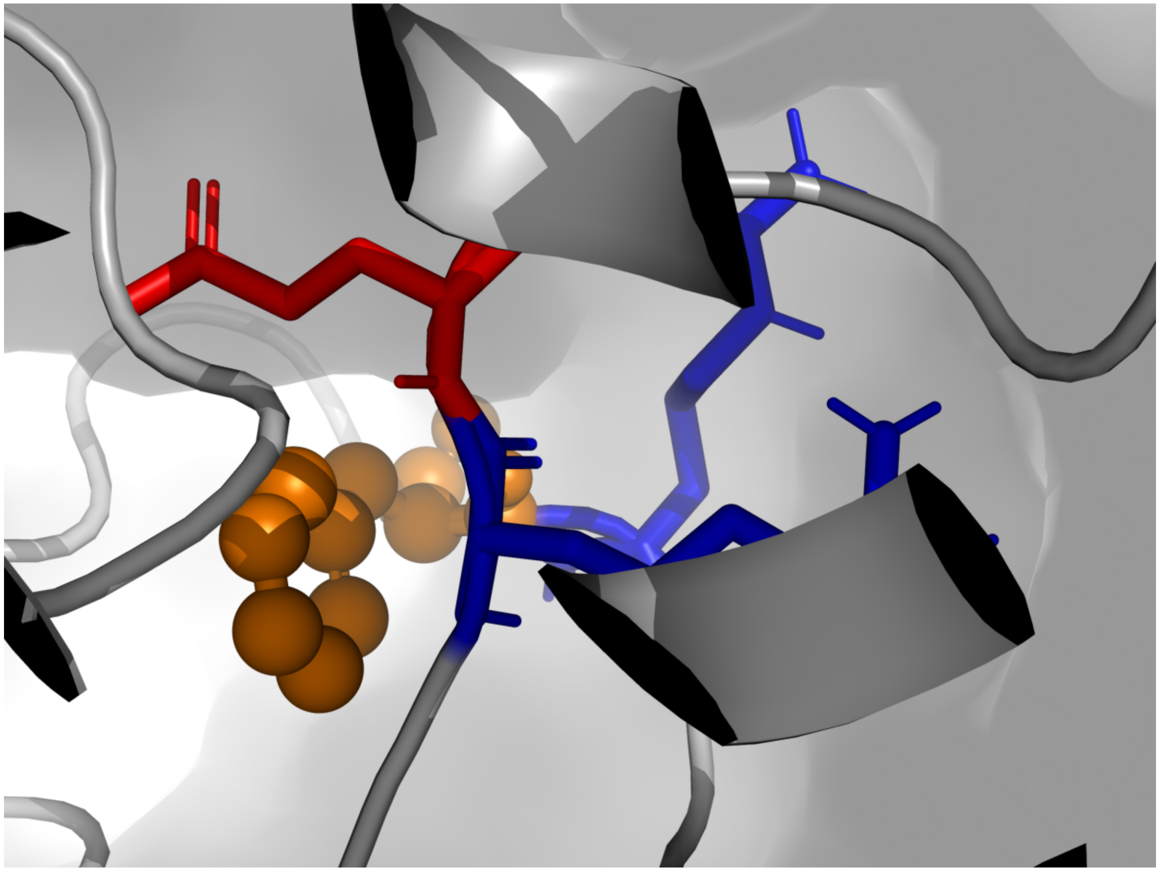
Solution:

- Shift paradigm to Thermodynamic Misfolding Severity (ddG).
- Identify mutations that physically tear the protein apart to create binding pockets.

Slide 2: Mutational Landscape (ddG vs SASA)



Slide 3: The C422F Neo-Pocket Discovery



Target Terrain Features:

• Center: F422 (Hydrophobic)

• Left: Arg421 / Arg433 (+)

• Right: Glu434 (-)

WT SASA = 0.0 A^2

Mutant SASA = 20.49 A^2

Perfect Druggable Neo-Pocket!

Slide 4: Pharmacophore Design (The Clamp)

Molecular Clamp Strategy

Left Anchor (-)
Carboxylic Acid
Binds Arg421 (+)

Core (Aromatic)
Aromatic Ring
Pi-Pi Stacking F422

Right Anchor (+)
Amine Group
Binds Glu434 (-)

Conclusion: Synthesize a bis-aryl scaffold with carboxylate and amine termini

to forcefully clamp the C422F structural defect and restore IDS stability.